

ing, flexo printing, screen printing, slot die coating and, most recently, ink-jet printing—are compatible with organic semiconductor processing. The printing solvent is evaporated when heated to moderate temperatures, producing a dried layer of the photoactive polymer. The modules are encapsulated between thin, flexible over-laminates, which protect active layers from mechanical abrasion and the environment. The advantages are low investment, printing and coating at a high speed, and no limitation on the substrate width. The processing steps are summarised in Fig. 2.

Thin, lightweight and flexible organic PVs are well suited for use on the outsides of buildings and on irregularly shaped products, such as fabrics for backpacks and tents. Several companies are actively pursuing these applications of organic PVs. Prominent among them are Solarte Belectric, Heliatak, Eight19, epishine, SolarWindow, Armor Beautiful Light, Raynergy Tek, Next Energy Technologies, Solarmer and Sunew Brasil. Dresden, Germany-based Heliatak, for example, has several pilot projects under way that highlight their ability to integrate foil-like organic photovoltaic modules into building facades made of glass, concrete and metal.

Heliatak has photovoltaic module production capacity of 10,000–20,000 sq. metres per year and their mass production target year is 2018. The other leading manufacturers of organic photovoltaics are Merck, Sumitomo, Mitsubishi and infinityPV.

InfinityPV sells solar cells and modules specifically for educational purpose, testing and manufacturing



Fig. 3: Thin organic solar-cell panels can be integrated into cement and metal portions, not just glass, of a building facade (Courtesy: Heliatak)

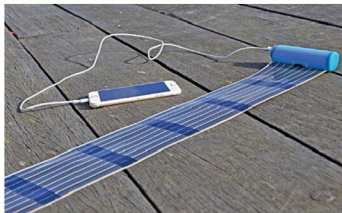


Fig. 4: Unroll this thin, flexible organic photovoltaic panel and charge your phone or battery indoors or outdoors, then roll it back up for storage (Courtesy: infinityPV)

analysis. In 2013, it developed a large-scale roll-to-roll printing technique and record of connecting 16,000 organic solar cells in series. InfinityPV also makes solar chargers for phones. The device features a hand-sized case with a retractable organic photovoltaic panel and a built-in lithium-ion battery.

The key hurdle in the commer-

cialisation of organic photovoltaics is financial, not technical. Organic PV cells' low-cost advantage can be attained only when components and modules are produced at a much larger scale of at least one million sq.m per year, which requires a high investment, and pilot-scale demonstrations to boost investors' confidence.

Dye-sensitised solar cells

Dye-sensitised solar cells can produce electricity in different light conditions, indoors and outdoors, enabling users to convert both artificial and natural light into energy to power a broad range of electronic devices.

All photovoltaic devices generate electricity through a series of light absorption, electronic excitation and charge separation events. In dye-sensitised solar cells, these phenomena occur in separate molecular layers, as depicted in Fig. 5.

Dye-sensitised solar cells have a porous network of disordered titanium dioxide nanoparticles that are coated with light-harvesting dye molecules and are typically surrounded by a liquid-phase electrolyte. When light beam passes through the transparent electrode of a dye-sensitised solar cell, it is absorbed by a dye (red), generally a ruthenium complex which coats with ZnO or TiO₂ nanoparticles (gray). The process creates photo-induced pairs of negatively charged electrons and positively charged holes (e^-/h^+). The charges separate at the surface of the nanoparticles: Electrons are injected into and transported through the ZnO or TiO₂ layer to one electrode, and positive charges migrate via the electrolyte to the opposite side of the cell, generating electric current.